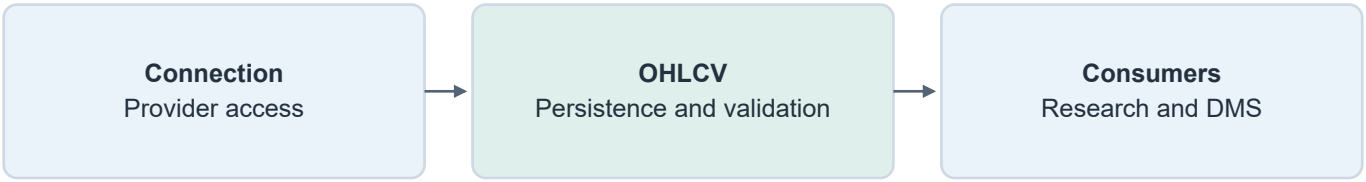
Connection Suite - Complete Map

LEONARDO V2

Scope, architecture, workflows, lifecycles and integration of the current Connection implementation. Targeted 28 October 2026 companion revision for accepted R01-R06. The 9 October edition evidence remains historical.

Still in development
Connection Suite is incomplete and will undergo further development. Its structure can still change significantly. This is a current-state map, not a frozen future design.



One reference set

Written manual	Use
CONNECTION_SUITE.md	Purpose, implemented scope, daily workflow and known limits
CONNECTION_ARCHITECTURE.md	Production composition, authority, dependencies and reconstruction
CONNECTION_LIFECYCLES.md	Ordered operations, mutation, failures and surviving state
CONNECTION_DATA_CONTRACTS.md	Algorithms, wire/data formats, exact fields, APIs and verification

Additive visual companion
The map supplements all four manuals. "Complete Map" refers to coverage of the current documented implementation; it does not mean the suite is a completed product.

This edition changes documentation only. Existing Core, OHLCV, Research and DMS authorities remain the reconstruction boundary. No future account, websocket or trading design is prescribed.

Implemented scope and unfinished surfaces

Implemented now	Actual behavior
Public historical REST	Default Bybit adapter; spot, linear, inverse; standard klines
Historical Download Manager	Capability-derived choices, preflight, confirmation, progress, Stop and recap
Core-supervised work	Task identity, status, cancellation, audit and coarse connection summaries
OHLCV handoff	Store writes and preliminary checks; Maintenance owns final validation

Presentation does not imply capability

Unfinished surface	Current limit
Connection home tables	Provider/account, websocket and download-overview tables are not live read models
Download stage/output tables	Shell surfaces remain empty; summary, progress, log and recap are wired
Private / live integration	No credentials, accounts, private APIs, live subscriptions, reconnect or order gateway
Queue and providers	No persisted download queue or automatic resume; only Bybit registered by default

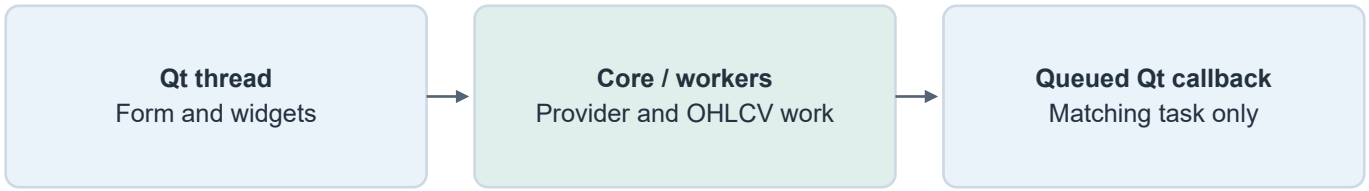
Future structure remains open
Further work may significantly change Connection internals. That statement is not approval to change shared ownership or the established data lifecycle.

Supported UI timeframes: 1m, 3m, 5m, 15m, 30m, 1h, 2h, 4h, 6h, 12h, 1d, 1w, 1M. Standard Bybit klines reject options. A syntactically valid symbol is not proof of exchange availability.

Active production composition



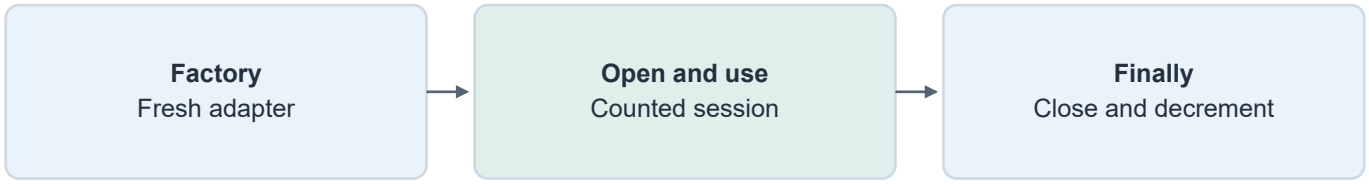
Composition object	Receives / owns
ConnectionApplicationService	Provider factory registry + Core ConnectionRegistry; scoped sessions
HistoricalDownloadService	Connection service + Store + audit + shared dataset locks
Download application service	Shared CoreRunner + domain downloader
Maintenance application service	CoreRunner + Maintenance + downloader + same dataset locks; loader invalidator
Accepted catalog / loader	Read-side access to the same configured historical root
HistoricalDownloadPresenter	Connection/download application services and DMS evidence-change callback



Reuse the existing runtime
No separate executor, provider singleton, child process or persistence authority is introduced. Blocking operations use existing workers; windows do not call provider APIs or write files.

Core/app is the composition root; provider and dataset meaning remain with their Areas. Object IDs on presenter-owned dialogs do not imply independent tracked Core window rows.

Provider sessions, discovery and retries



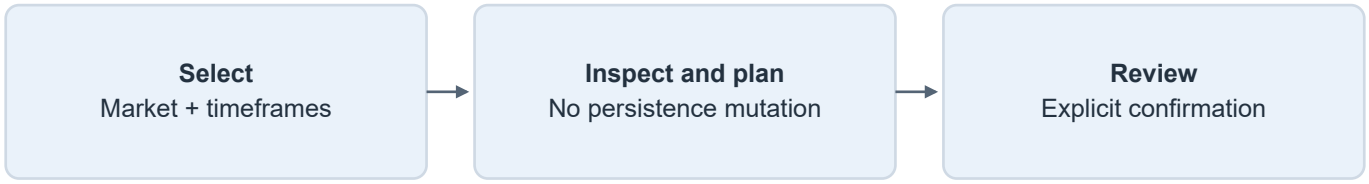
Provider boundary	Current contract
Default access	Bybit public v5 /market/time and /market/kline; mainnet default
Capabilities	Static queries instantiate an adapter without opening HTTP
Connection row	connection.bybit.historical_rest; connected until clean last exit
Concurrent failure	Coarse degraded with other sessions; failed for sole failed session
Candle conversion	Numeric timestamp/OHLCV, ascending rows; newest open flag when server-time evidence permits
Oldest discovery	Binary existence probes; final bounded fetch uses 1,000 regardless of selected data-page limit

Bounded transport policy

Layer	Limits
Adapter	30-second default HTTP timeout; 0.20-second pacing per instance; up to four attempts
Rate limiting	HTTP 429 / code 10006; reset-header delay or fallback; maximum 10-second sleep
Downloader page wrapper	30-second wait; up to three attempts; typed permanent/already-exhausted failures stop outer retry

Two retry layers, explicit meaning
Planning calls use adapter behavior. Page calls add a downloader timeout. An outer timeout may occur before the adapter exhausts its attempts; this is not one global four-request budget.

Request, preflight and plan



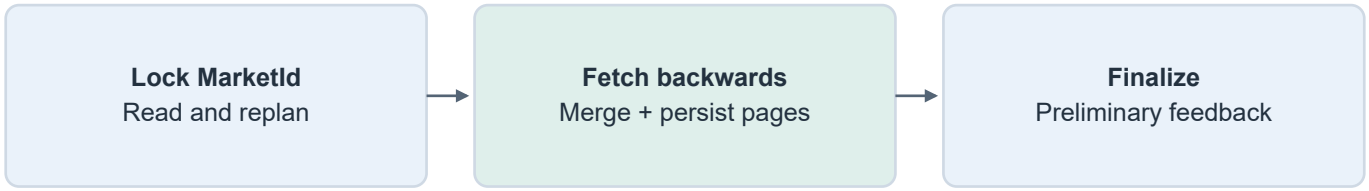
Plan / flag	Inputs and range
new_download	No custom boundaries and no local rows: provider oldest to latest closed
update_latest	Automatic existing dataset: local last timestamp overlaps latest tail
custom_range	Start, End or both supplied; inclusive filtering and merge
up_to_date flag	Only automatic update with known bounds and local last >= latest closed

Preflight is a snapshot
Local files and provider time can change after review. Execution rereads and replans under the dataset lock. Preflight neither reserves files nor grants Research acceptance.

Review rule	Meaning
Automatic new history	Blocked if the oldest provider timestamp cannot be resolved
Expected bars/pages	Fixed-duration estimate; unknown bounds and calendar months can be indeterminate
Limit	Blank/zero uses provider default; nonzero capped at provider maximum
Warnings	Local issues remain visible; confirmable does not mean all later I/O will succeed

No Start/End automatic tail update does not repair internal gaps or recover an earlier missing prefix. Custom ranges retain their explicit-boundary semantics.

Execution and the true no-op branch



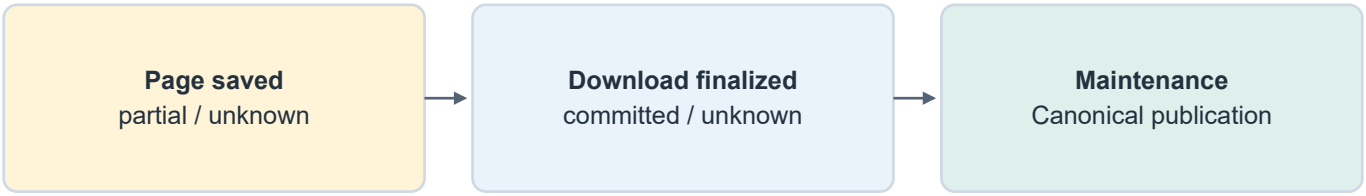
Step	Observable contract
Page conversion	Sort, apply bounds, automatic first-page open-candle exclusion
Merge	Incoming wins at equal timestamps; preserve other rows; ascending CSV
Persist	Each page saves merged dataset as partial / unknown, then emits page_persisted
Next cursor	Oldest filtered timestamp minus one millisecond
Normal end	Known lower bound reached, empty response/filter, or negative cursor
Ceiling	10,000 pages exhausted without a break: fail before finalization
Normal finalize	Nonempty dataset; preliminary report; committed / unknown

True up-to-date update
Run read-only preliminary validation and return up_to_date=True. CSV, sidecar, mtimes, provenance and validation evidence remain unchanged. No mutation notification is sent.

Zero fetched rows is a different statement
A custom request with no new rows may still finalize existing data and reset acceptance. Missing, invalid, stale or partial evidence is never promoted by the true no-op branch.

Persistence and admission to trusted use

One canonical dataset: historical root / exchange / market type / symbol / timeframe segment / ohlcv. Files: candles.csv and candles.meta.json. Month 1M uses directory 1mo; minute uses 1m.



Evidence / state	Authority and checks
CSV payload	Exactly ts_ms, open, high, low, close, volume; Store owns writes
OHLCVSidecarV1	Schema 1.0; MarketId, hash/count/range, source, states, warnings, lineage and UTC times
Preliminary validation	Read-only defensive feedback; never publishes acceptance
Canonical publication	Stable CSV and sidecar size/mtime/SHA; final persistence; report and evidence agree
Strict accepted catalog	Canonical identity + final persistence + validation ok + nonempty data + current matching CSV hash

Atomic per file, not across the pair
CSV and sidecar are replaced separately. An I/O failure can leave unmatched evidence. Current consumers must refuse stale evidence; task text cannot override it.

Canonical quality can be warning or error and still be publishable without being accepted. Metadata-only discovery is not payload verification. A true no-op preserves whichever prior accepted or rejected state existed.

Cancellation, failures and surviving data

Cancellation or failure point	What survives
Before a new successful page	Prior durable state; no new page is implied
During page write	Shielded write settles; successful page is reported, then cancellation propagates
Between pages / ceiling failure	Saved rows remain partial / unknown; no rollback or automatic resume
No-op preliminary check	Previously stored CSV and validation evidence stay unchanged
During finalization	Finalization settles; task can be cancelled after committed / unknown evidence exists
Later timeframe fails	Earlier completed timeframes remain persisted
Filesystem failure	Inspect actual file/evidence state; no blanket transaction guarantee

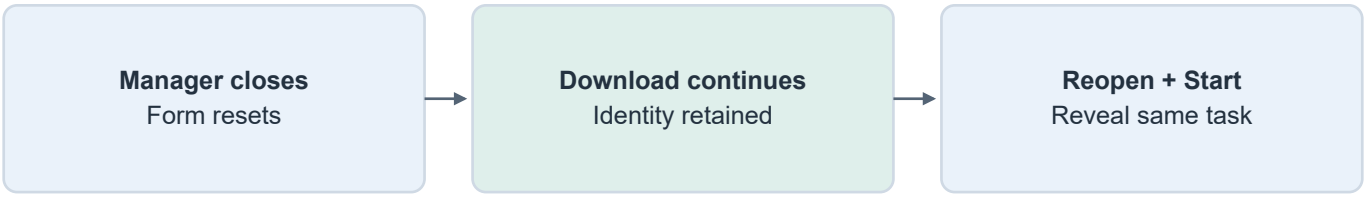


Current recap limitation
The GUI uses the last item-cancel event, not a complete batch cancellation report. Its zero-page text can say validation is unknown even when a cancelled no-op preserved acceptance.

Item and batch events/audit retain additional cancellation evidence. The underlying historical Windows/native abort and intermittent CSV replacement causes are not claimed to have been identified. Existing reviewed Maintenance recovery remains the supported route where eligible.

Task identity and window lifetime

Object	Current lifetime rule
Manager form	Close resets input, cancels pending preflight, retains active download
Preflight dialog	Own task ID; close clears review; late results ignored
Download task dialog	Retained by presenter; closing display is not Stop
Confirmed request	Separate from editable/reset form until active download settles
Active task ID	Both start entry points reveal existing job; stale progress/results ignored before side effects
Terminal result	Stop timer, flush pending display, clear active ownership once; enable OK



Notification before display throttling
First successful page persistence notifies an open DMS context once per MarketId per job. Matching events may then be coalesced for the 250 ms display throttle.

Tracked long-lived IDs: connection_suite.home.window, historical_download_manager.window and ohlcv_maintenance.window. Preflight/task dialog object IDs remain presenter-owned and are not separate tracked window rows in current composition.

Progress percentages and GUI logs are disposable views. They do not replace task status, dataset evidence or a durable resumable job.

Maintenance and controlled recovery

Operation	Review, mutation and result
Discover	Read-only listing; problematic datasets may be visible even when Research rejects them
Validate	Cooperative worker scan; stable evidence; optimistic Store publication; accepted only when published ok
Repair	Derive safe ranges; explicit execution; stale-plan check under dataset lock; redownload reviewed ranges; mark repaired then validate
Reconstruct sidecar	Eligible stable CSV/evidence only; confirm; create committed / unknown then validate; invalidate loader cache
Delete	Review exact CSV/optional sidecar fingerprints; confirm; stage files, remove, clean only empty canonical directories; invalidate cache

Locks have defined boundaries
Download, repair ranges, deletion and reconstruction share MarketId locks. Ordinary validation uses stable evidence and optimistic publication. Repair retains its lock through repaired marking, post-validation and result construction. Cancellation drains the finalizer before release; this is not a crash transaction.

No blanket promotion or cleanup
Partial evidence cannot become committed through sidecar reconstruction. Unsupported structural repair remains inspect-only. Deletion does not cascade to saved Research or DMS object pools.

Confirmed deletion/reconstruction is shielded once started and the presenter does not offer cancellation then. Late cancellation is reported alongside the actual result. Staging failure attempts restoration; later cleanup failures remain explicit warnings.

Leonardo integration and shutdown



Neighbor	What Connection does and does not own
Research	Accepted catalog/loader are the handoff; no push refresh of open charts or automatic study recalculation
DMS	Composition forwards evidence change and accepted-validation callbacks; existing ownership/compatibility remain
Saved objects	Source deletion leaves definitions/references with their owners; it does not guarantee those references remain usable
Loader cache	Explicit invalidation in deletion/reconstruction; do not claim every successful operation calls it
Runtime Manager	Generic tasks/connections from Core; no historical-download child process

Application exit uses existing Core lifecycle



Provider contexts close through task unwinding, not by changing a registry row. Core shutdown may fail if work cannot settle within the timeout. Restart composes new runtime objects; CSV/sidecar state survives, form and task display do not. No automatic resume or acceptance is inferred.

Reconstruction, evidence and development status

Rebuild in this order	Use the detailed manuals
1. Identity and provider port	MarketId, sidecar, ProviderCandle, registry, scoped sessions and Bybit adapter
2. OHLCV behavior	Store, validation, dataset locks, planning, paging and cancellation boundaries
3. Application integration	Existing Core submissions and Maintenance review/publication paths
4. GUI and consumers	Production composition, retained task identity, Qt dispatch and accepted-data handoff
5. Verification	Source-exact inventory: 32 dataclasses and 93 public interfaces; behavior test matrix

Evidence is bounded and dated

Record	What was established
9 October implementation	2,890 tests passed; 7 Windows symlink-privilege skips; 8 generated-data native checks
9 October independent audit	225 fresh passing checks; unchanged full/native records verified, not rerun
28 October revision	R01-R06 source reconciliation; document/PDF checks only. Original counts above are historical; R08/R09 remain pending
Limits	No live account/exchange acceptance; historical native fault mechanism unresolved; user visual acceptance separate

Incomplete; further development expected
The Connection structure can still change significantly. These documents explain what exists today and where its authority stops. They neither freeze future internals nor approve new product behavior.

Original edition evidence: Leo_V2_Task_Packages / 20261009_Connection_Documentation. R07 revision: 20261028_T1102_R07_R1. All four manuals remain the detailed implementation reference; use this map to navigate them.